Heating System – General Data

1. General

The heating system utilizes heated and compressed bleed air from tappings on both engine compressors, mixes the bleed air with secondary air (ram, fan pressure, or recirculated air) and distributes it to outlets in the cabin and flight compartment.

2. Description

Refer to Figure 1 for details.

The heating system consists of an interconnected bleed pipe assembly incorporating two solenoid-operated shut-off valves and two check valves, hot air valve, expansion chamber, ejector, silencer, cabin air control valve, ducts and outlets. For system control refer to 21-60-00, Environmental Control System (Mod 6/2014) – General Data.

A. Bleed Air Pipe Assembly

The pipes from each engine compressor casing are routed to their respective shut-off valves, (located in the engine nacelles), and from the shut-off valves through the wings to join a tee pipe in the cabin roof; check valves are installed in each end of the tee pipe. From the tee pipe, a common pipeline routed forward, above the cabin ceiling and down the rear left-hand side of the cabin/flight compartment bulkhead, to connect to the hot air valve under the flight compartment floor. A “De Vore” protection is installed over the bleed air pipes from the shut-off valves through the wings to the cabin roof. Pipes are insulated with fiberglass and are sealed to exclude fluids.

B. Hot Air Valve

The hot air valve, connected between the bleed air pipe assembly and a pipe connected to the expansion chamber, is located below the pilot's seat under the flight compartment floor. The valve is of the motorized butterfly type which takes approximately 30 seconds to move through its full 90 degree range of travel, thus providing control sensitivity.

C. Expansion Chamber

The expansion chamber, which reduces bleed air noise, is a duct into which protrudes a pipe connected to the inlet end of the duct; the pipe from the hot air valve connects to this end. The outlet end of the expansion chamber incorporates a group of nozzles which protrude into the ejector and a flange which bolts to the ejector.

D. Ejector

The ejector, located laterally across the fuselage below the flight compartment floor, forms the mixing chamber for recirculated air. The nozzles, protruding into the ejector and the ejector diffuser, induce a suction which draws in recirculated air from the flight compartment when ram or fan pressure air is not being used as the secondary air source. The recirculated air duct, connected between an intake in the flight compartment footwell and the rear of the ejector, incorporates an integral silencer and check valve.

E. Silencer

The silencer is located longitudinally below the flight compartment, its bifurcated inlet connects to the ram air fan duct and ejector outlet. The silencer outlet, which incorporates the cabin air control valve, connects to the distribution ducting. Two smaller diameter outlets on the silencer provide for the connection of flight compartment, windshield and foot warmer ducts. Provision is also made on the silencer for the installation of temperature control system components.

F. Cabin Air Control Valve

The cabin air control valve is located in the silencer outlet duct and controls the supply of warm (with hot air valve and engine bleed valves closed) or the cooling air engine supply to the cabin. The valve is manually operated by a handle located in the cockpit central pedestal. Closing the air control (by pulling the handle fully out), allows all available flow to be directed into the flight compartment.

G. Ducts

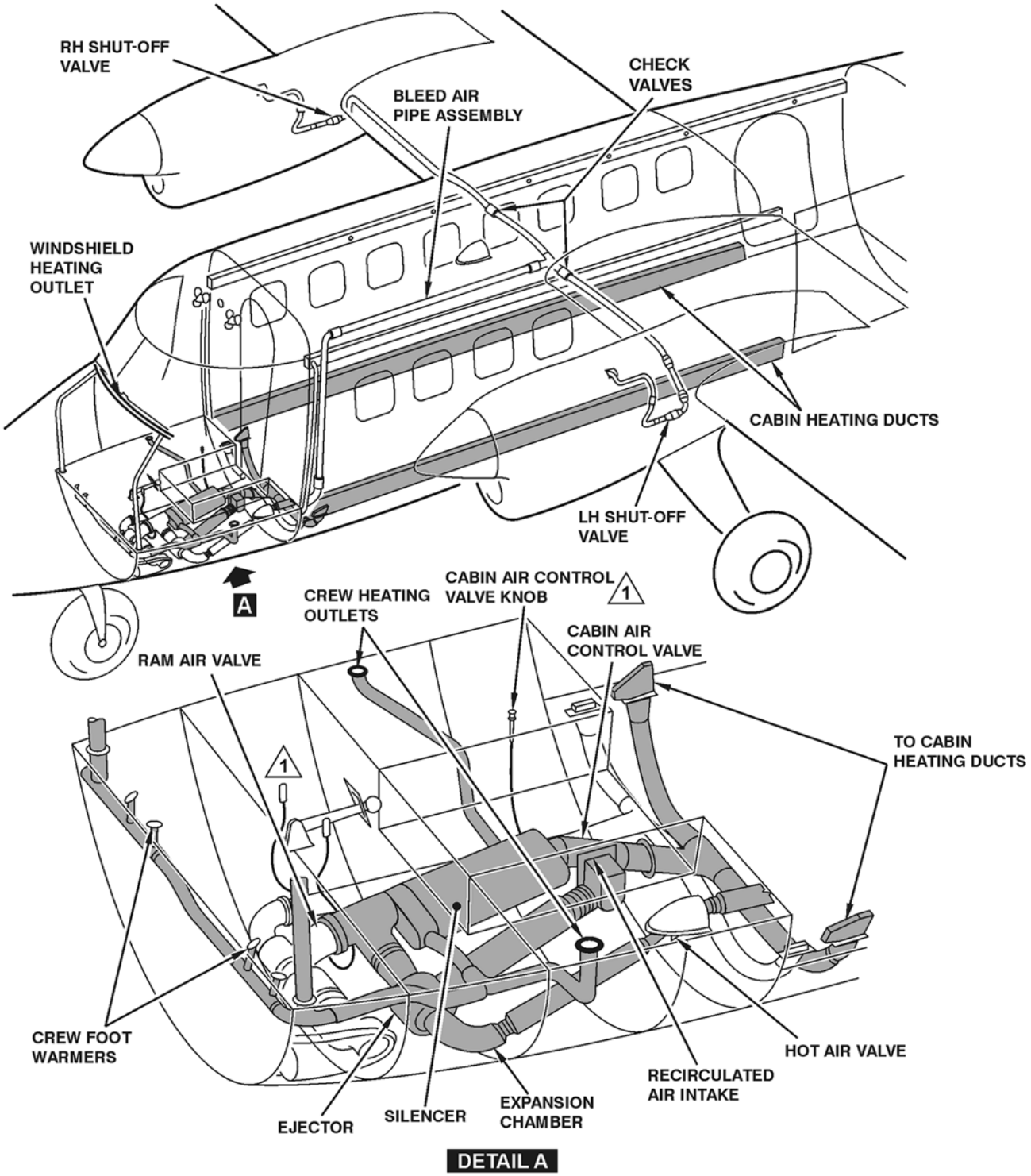
The heating system ducts comprise an installation of aluminum alloy and polycarbonate tubes which lead from the silencer to the pilot's and co-pilot's foot warmers, the windshield heater and cabin heater outlets.

H. Outlets

The heating system outlets connect to the terminating ends of the windshield, crew and cabin heating system ducts. The outlets are of polycarbonate or nylon material and formed to provide a suitably diffused air flow. The windshield outlets are integrated with the glare shield above the instrument panel, and the crew foot warmer outlets are at floor level and direct their flow towards the rudder pedals. The cabin outlet ducts are box section ducts which extend from the flight compartment/cabin bulkhead rearwards at the base of each wall.

I. Cabin Temperature Indication

The cabin air temperature will be displayed through CAS on the environmental panel (refer to 21-60-00, Environmental Control System (Mod 6/2014) – General Data).



Heating System Installation
Figure 1

3. Operation

With the engines running, bleed air from the engine compressors is controlled by the solenoid-operated shut-off valves which are energized by selecting switches on the co-pilot panel. The switches are identified BLEED AIR, LEFT and RIGHT, and since they are normally ON bleed air from the shut-off valves passes through the check valves, which prevent reverse flow in the event of unequal engine compressor casing pressures, and through the common supply pipeline to the hot air valve. The hot air valve, receiving signals from the temperature control system, opens or closes to regulate the amount of bleed air into the ejector. A flow limiting orifice, which limits the bleed air flow to 4 percent of the two-engine bleed flow-rate, is located immediately upstream of the hot air valve. Bleed air noise is reduced by the expansion chamber just prior to entering the ejector.

Depending on temperature selection, position of ram air valve handle, and if ram air or fan pressure air is available, the ejector will either draw in recirculated air for mixing with the bleed air, or eject hot air to mix with ram or fan air before entering the silencer. With the cabin air control valve open, the main flow of warm air will pass through the cabin ducts to the cabin outlets, where it is distributed through a pattern of round perforations in the inward facing side of each outlet duct. A limited supply of warm air is provided to the flight compartment, foot warmer and windshield heating outlets. If, however, the cabin air control valve is closed, all available warm air will be distributed to the latter and the supply to the cabin outlets is shut-off.

4. Adjustment/Test

A. Test Heating System Bleed Air Pipe Installation

Equipment required:

- Source of compressed air 100 psi
- Pressure test rig consisting of a pressure tank capable of withstanding 160 psi and having a volume of 10 – 12 cu. ft., a 0 – 100 psi pressure gauge, and two shut-off valves to isolate the rig from the aircraft and the compressed air source.

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Set appropriate circuit breakers for system under test.

3) If a heat exchanger package is fitted, remove cabin roof upholstery in location of station 177.00 to provide access to bleed air system pipeline. Disconnect heat exchanger pipe from bleed pipe and blank off bleed pipe tapping.

4) Disconnect and blank off bleed pipe at first or second coupling downstream from engine (between take-off point and shut-off valve) in left nacelle, using a Marman style blanking disc.

WARNING BEFORE ATTEMPTING TO DISCONNECT PIPING, RELEASE ALL AIR PRESSURE FROM THE SYSTEM.

Note A suitable blanking disc 1.710 inches diameter x 0.065 ± 0.015 inch thick, may be manufactured from aluminum alloy.

5) Disconnect the left intake deflector pipe and connect test rig pressure line to bleed air pipeline.

6) Select LEFT, BLEED AIR switch ON and HEATING MODE switch to OFF. Apply a pressure of 40 psi and check all connections for obvious leaks. Rectify leaks as necessary.

Note Air will be leaking from bleed shut-off valve vents, this is normal and acceptable. If leak rate is high it may be necessary to replace valve to meet the requirements of Para 4.A.7).

7) Build up pressure to 80 psi and close shut-off valve on test rig to isolate rig from compressed air source; record time taken for system pressure to fall to 70 psi. If time taken is:

a) 2 minutes or more, system is good.

b) 30 seconds to 2 minutes, system is acceptable but at the earliest opportunity leaks must be rectified.

c) 30 seconds or less, system is unserviceable and leaks must be rectified immediately.

8) Select LEFT, BLEED AIR switch OFF, open hot air valve by selecting warm, and HEATING MODE switch to ON. Pressurize system to 70 psi.

9) Select LEFT, BLEED AIR switch ON and note that test rig pressure drops rapidly.

10) Disconnect test rig, select HEATING MODE switch to OFF and ensure LEFT, BLEED AIR switch is at ON.

11) Reconnect left intake deflector pipe to bleed air pipe. Remove blank and reconnect bleed air pipe coupling (between take-off point and shut-off valve).

12) Repeat Para 4.A.4) through Para 4.A.11) for right-hand side, using RIGHT, instead of LEFT, BLEED AIR switch.

13) If a heat exchanger package is fitted, remove blank and connect heat exchanger pipe to bleed pipeline tapping. Replace cabin roof upholstery.

14) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

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